

CRK, MTR

Installation and operating instructions



English (GB) Installation and operating instructions

Original installation and operating instructions

These installation and operating instructions describe Grundfos CRK and MTR pumps.

Sections 1-4 give the information necessary to be able to unpack, install and start up the product in a safe way.

Sections 5-11 give important information about the product, as well as information on service, fault finding and disposal of the product.

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Read this document before installing the product. Installation and operation must comply with local regulations and accepted codes of good practice.

1. General information

1.1 Hazard statements

The symbols and hazard statements below may appear in Grundfos installation and operating instructions, safety instructions and service instructions.

DANGER



Indicates a hazardous situation which, if not avoided, will result in death or serious personal injury.

WARNING



Indicates a hazardous situation which, if not avoided, could result in death or serious personal injury.

CAUTION



Indicates a hazardous situation which, if not avoided, could result in minor or moderate personal injury.

The hazard statements are structured in the following way:

SIGNAL WORD



Description of hazard

Consequence of ignoring the warning.
- Action to avoid the hazard.

1.2 Notes

The symbols and notes below may appear in Grundfos installation and operating instructions, safety instructions and service instructions.



Observe these instructions for explosion-proof products.



A blue or grey circle with a white graphical symbol indicates that an action must be taken.



A red or grey circle with a diagonal bar, possibly with a black graphical symbol, indicates that an action must not be taken or must be stopped.



If these instructions are not observed, it may result in malfunction or damage to the equipment.



Tips and advice that make the work easier.

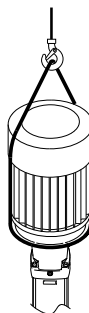
2. Receiving the product

2.1 Lifting the product

When lifting the entire product with motor, follow these instructions:

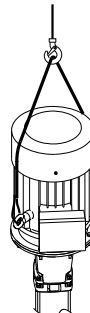
- Pump with motor sizes 0.37 - 5.5 kW:
Lift the pump in the motor flange by means of straps or the like.
- Pump with motor sizes 7.5 - 22 kW:
Lift the pump by means of the motor eyebolts.
- Pump with motor sizes 30-45 kW:
Lift the pump by means of the lifting brackets on the motor flange.

0.37 - 5.5 kW



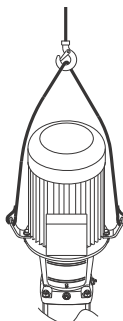
TM04 0339 0608

7.5 - 22 kW



TM04 0341 0608

30-45 kW



TM05 9564 4113

55-75 kW

Fig. 1 Lifting points

In case of pumps with other motors than MG or Siemens, we recommend that you lift the pump by means of the straps in the motor flange.

3. Installing the product

3.1 Mechanical installation

WARNING

Moving parts

- Death or serious personal injury
- Tighten the coupling screws to the correct torque.
 - Install the coupling guards securely to the pump with the screws intended for this purpose.

CAUTION

Hot surface

- Minor or moderate personal injury
- The pump must be installed so that persons cannot accidentally come into contact with the hot surface of the motor.

3.2 Pump location

The pump is designed for tank mounting in vertical and horizontal position. However, only MTR version H and MTRE version H are suitable for horizontal mounting.

The pump is positioned in a hole cut into the tank and is secured to the tank by four hexagon head screws through the holes in the mounting flange.

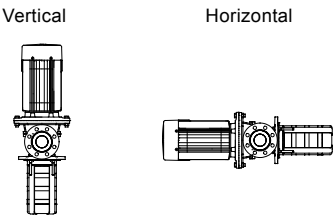


Fig. 2 Vertical and horizontal installation

On horizontally installed MTR, MTRE pumps with motors from 5.5 kW and up the motors have feet and must be supported.

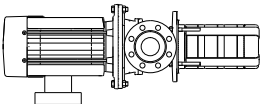


Fig. 3 Horizontally installed MTR with foot

For horizontally mounted pumps ensure a sufficient sealing between pump flange and tank. A sealing gasket will usually suffice.

Before changing shaft seals for horizontally mounted pumps, drain the tank.

MTR 32, 45 and 64 pumps can only be installed in vertical position.

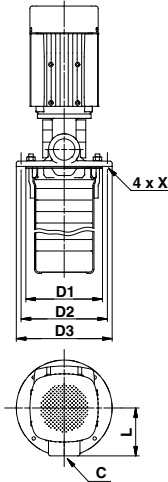


Fig. 4 Vertical installation

Mounting flange dimensions:

Pump type	D1	D2	D3	L	C	X
CRK 2, 4	140	160	180	100	Rp 1 1/4 G 1 1/4	Ø7.5
MTR 1s, 1, 3, 5, 8	140	160	180	100	Rp 1 1/4 G 1 1/4	Ø9.5
MTR 10, 15, 20	200	225	250	125	Rp 2 G 2	Ø9
MTR 32	190	220	250	150	DN 65	Ø12
MTR 45, 64	240	265	290	165	DN 80	Ø12

TM02 8042 4503

TM01 4990 1399

TM04 5755 3809

3.2.1 Inlet conditions

The bottom of the pump strainer must be at least 25 mm above the bottom of the tank.

The pumps are designed to provide full performance down to a level of A mm above the bottom of the strainer.

At a liquid level between A and B mm above the bottom of the strainer, the built-in priming screw will protect the pump against dry running.

MTR 32, 45 and 64 pumps have no priming screw.

Pump type	A [mm]	B [mm]
CRK 2, 4 and MTR 1s, 1, 3, 5	41	28
MTR 10, 15, 20	50	25
MTR 32, 45, 64	70	-

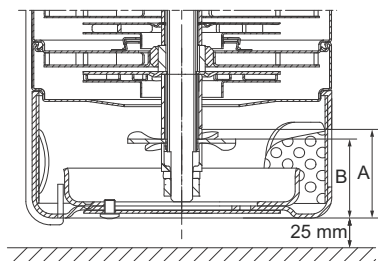


Fig. 5 CRK 2, 4 and MTR 1s, 1, 3, 5, 8

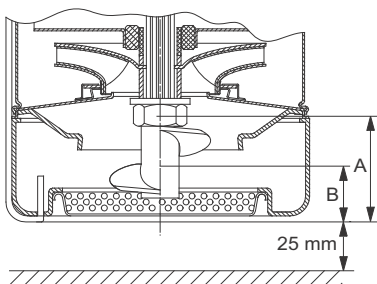


Fig. 6 MTR 10, 15, 20

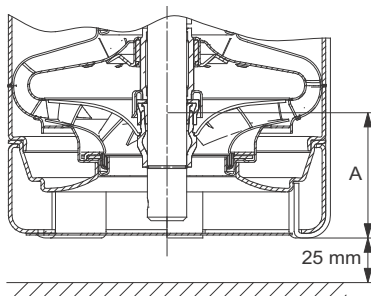


Fig. 7 MTR 32, 45, 64

3.3 Electrical connection

WARNING

Electric shock

Death or serious personal injury

- Before starting any work on the product, make sure that the power supply has been switched off and that it cannot be accidentally switched on.
- Connect the pump to an external main switch close to the pump and to a motor-protective circuit breaker or a CUE frequency converter. Make sure that you can lock the main switch in OFF position (isolated). Type and requirements as specified in EN 60204-1, 5.3.2.



The electrical connection must be carried out by an authorized electrician in accordance with local regulations.

The electrical connection must be carried out as shown in the wiring diagram inside the terminal box cover.

The operating voltage and frequency are marked on the pump nameplate. Please make sure that the motor is suitable for the power supply on which it will be used.

Single-phase Grundfos motors incorporate a thermal switch and require no additional motor protection.

TM05 7225 0813

TM07 2800 4118

TM05 7224 0813

Three-phase motors must be connected to a motor-protective circuit breaker.

The terminal box can be turned to four positions, in 90 ° steps. See fig. 8.

1. If necessary, remove the coupling guards. Do *not* remove the coupling.
2. Remove the bolts securing the motor to the pump.
3. Turn the motor to the required position.
4. Replace and tighten the bolts.
5. Replace the coupling guards.

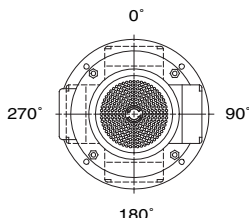


Fig. 8 Terminal box positions

3.3.1 Frequency converter operation

Motors supplied by Grundfos

All three-phase motors supplied by Grundfos can be connected to a frequency converter.

Depending on the frequency converter type, this may cause increased acoustic noise from the motor.

Furthermore, it may cause the motor to be exposed to detrimental voltage peaks.



Grundfos motors, types MG 71 and MG 80 as well as MG 90 (1.5 kW, 2-pole), for supply voltages up to and including 440 V (see motor nameplate), must be protected against voltage peaks higher than 650 V (peak value) between the supply terminals.

We recommend to protect all other motors against voltage peaks higher than 850 V.

The above disturbances, i.e. both increased acoustic noise and detrimental voltage peaks, can be eliminated by fitting an LC filter between the frequency converter and the motor.

For further information, please contact the frequency converter or motor supplier.

Other motor makes

If other motors makes than those supplied by Grundfos are used, contact Grundfos or the motor manufacturer.

4. Starting up the product

WARNING

Too high pressure and leakage

Death or serious personal injury

- Do not run the pump against a closed outlet valve for more than approx. 5 minutes.



4.1 Before starting the product

- Make sure that all pipe connections are tight.
- Make sure that the pump is partly filled with liquid (partly submerged).
- Make sure that the strainer is not blocked by impurities.

4.2 Starting the product

1. Close the isolating valve on the outlet side of the pump.
2. If the pump is fitted with a vent valve, this valve must be opened. See fig. 9.

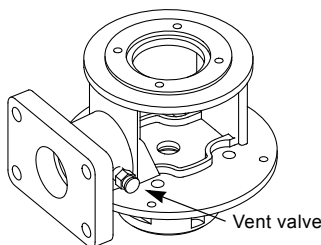


Fig. 9 Position of the vent valve

3. See the correct direction of rotation of the pump on the motor fan cover or on the coupling guard. When seen from the top, the pump must rotate counterclockwise.
 4. Start the pump and check the direction of rotation.
 5. Open the outlet pipe isolating valve a little.
 6. If the pump is fitted with a vent valve, this valve must be closed when a steady flow of liquid runs out of it.
 7. Completely open the outlet pipe isolating valve.
- The pump has now been vented and is ready for operation.

5. Product introduction

5.1 Applications

The Grundfos pumps, types CRK and MTR, are tank mounted multistage centrifugal pumps designed for the following applications:

- liquid transfer in machine tools
- condensate transfer
- liquid transfer in industrial washing machines
- pressure boosting of cold or hot clean liquids
- similar applications.

5.2 Pumped liquids

Thin, non-explosive liquids, not containing fibres. The liquid must not attack the pump materials chemically.

When pumping liquids with a density and/or viscosity higher than that of water, use motors with correspondingly higher outputs, if required.

MTR version A pumps have a cast iron pump head.

In MTR version I and CRK version I pumps all parts in contact with the pumped liquid are of stainless steel ENDIN 1.4301 or better.

5.3 Identification

5.3.1 Nameplate for CRK and MTR

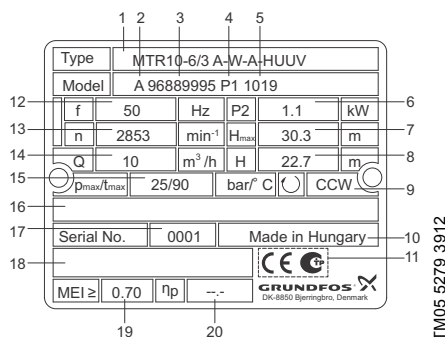


Fig. 1 Example of nameplate

Pos.	Description
1	Type designation
2	Model
3	Product number
4	Place of production
5	Production year and week (YYWW)
6	P2, 50 Hz
7	Head against closed valve
8	Head at rated flow rate
9	Direction of rotation CCW: Counterclockwise CW: Clockwise

Pos.	Description
10	Country of production
11	Approval marks
12	Frequency
13	Speed
14	Rated flow rate
15	Maximum pressure and temperature
16	The number of the copy of the technical file kept at KEMA (stated if the pump is ATEX classified)
17	The serial number of the pump (stated if the pump is ATEX classified)
18	ATEX category (stated if the pump is ATEX classified)
19	Minimum efficiency index
20	Hydraulic pump efficiency

5.3.2 Type key for CRK

Example CRK E 4 - 160 / 2 -x -x -x -xxxx

Pump type	
Pump with integrated frequency control	
Rated flow rate [m ³ /h]	
Number of chambers* x 10	
Number of impellers*	
Code for pump version	
Code for pipe connection	
Code for materials	
Code for shaft seal and rubber pump parts	

* See fig. 2.

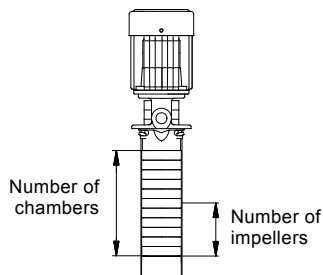


Fig. 2 Number of chambers / impellers

5.3.3 Type key for MTR, MTR E

Example	MTR	E	32	(s)	-2	/1	-1	-A	-F	-A	-H	UU	V
Pump type													
Pump with integrated frequency control													
Rated flow rate [m³/h]													
All impellers with reduced diameter (applies only to MTR 1s)													
Number of chambers, see fig. 2													
Number of impellers, see fig. 2													
Number of impellers with reduced diameter													
Pump version													
A Basic version													
B Oversize motor													
C Inlet pipe													
E Pump with certificate/approval													
F 120 °C version													
H Horizontal version													
J Pump with different max. speed													
P Undersize motor													
T Double oversize													
X Special version													
Pipe connection													
F DIN flange													
G ANSI flange													
J JIS flange													
M Square flange with internal thread													
W Internal thread													
WB NPT internal thread													
X Special version													
Materials													
A Basic version													
I Wetted parts EN/DIN 1.4301 / AISI 304													
X Special version													
Shaft seal													
H Balanced cartridge seal													
Q Silicon carbide													
U Tungsten carbide													
B Carbon													
E EPDM													
F FXM													
K FFKM													
V FKM													

6. Technical data

Pump type	CRK	MTR
Minimum liquid temperature [°C]	-10	-10
Maximum liquid temperature [°C]	+90	+90 (120*)
Maximum operating pressure [bar]	25	25
Enclosure class	IP55	IP55

* Applies to pump version F.

6.1 Ambient temperature

Motor power [kW]	Motor make	Motor efficiency class	Max. ambient temperature at full load [°C]	Max. altitude above sea level [m]	Pos. in output curve in fig.
0.06 - 0.18	Siemens	-	+40	1000	1
0.25 - 0.55	MG	-	+40	1000	1
0.75	MG	IE2	+60	3500	2
1.1 - 22	MG	IE2	+60	3500	2
30-45	Siemens	IE2	+55	2750	3

If the ambient temperature exceeds the above temperature values or the pump is installed at an altitude exceeding the above altitude values, the motor must not be fully loaded due to the risk of overheating. Overheating may result from excessive ambient temperatures or the low density and consequently low cooling effect of the air.

In such cases, it may be necessary to use a motor with a higher rated output.

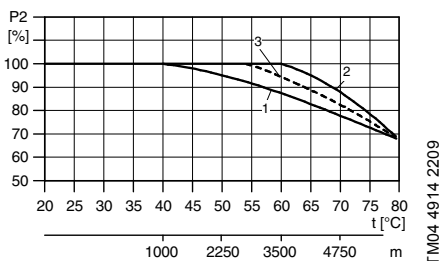


Fig. 3 Relationship between motor output (P2) and ambient temperature/altitude

Example: A pump with a 1.1 kW IE2 MG motor: If this pump is installed 4750 m above sea level, the motor must not be loaded more than 88 % of rated output. At an ambient temperature of 75 °C, the motor must not be loaded more than 78 % of rated output. If the pump is installed 4750 m above sea level at an ambient temperature of 75 °C, the motor must not be loaded more than 88 % × 78 % = 68.6 % of rated output.

6.2 Maximum permissible operating pressure and liquid temperature for the shaft seal

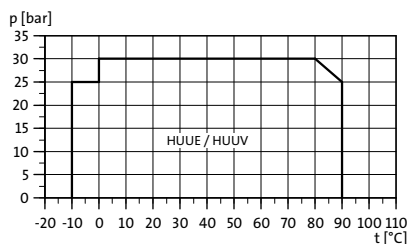


Fig. 4 MTR 1s to 64

6.3 Minimum flow rate

Due to the risk of overheating, the pump must not be used at flows below the minimum flow rate.

The curve below shows the minimum flow rate as a percentage of the nominal flow rate in relation to the liquid temperature.

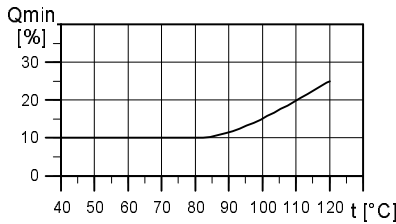


Fig. 5 Minimum flow rate in relation to temperature

6.4 Electrical data

See motor nameplate.

6.5 Maximum number of starts

Motor size [kW]	Recommended maximum number of starts per hour
0.06 - 0.18	100
0.25 - 2.2	250
3-4	100
5.5 - 11	50
15-22	40
30-45	8

7. Sound pressure level

The table below shows airborne noise emitted by CRK and MTR pumps with motors fitted by Grundfos.

Motor [kW]	L _{pA} [dB(A)]	
	50 Hz	60 Hz
0.37	50	55
0.55	50	53
0.75	50	54
1.1	52	57
1.5	54	59
2.2	54	59
3.0	55	60
4.0	62	66
5.5	60	65
7.5	60	65
11	60	65
15	60	65
18.5	60	65
22	66	70
30	71	75
37	71	75
45	71	75

8. Servicing the product

WARNING

Electric shock



- Death or serious personal injury
- Before starting any work on the product, make sure that the power supply has been switched off and that it cannot be accidentally switched on.

WARNING

Moving parts



- Death or serious personal injury
- Tighten the coupling screws to the correct torque.
 - Install the coupling guards securely to the pump with the screws intended for this purpose.

CAUTION



Corrosive liquids

- Death or serious personal injury
- Wear personal protective equipment.

CAUTION



Toxic liquids

- Death or serious personal injury
- Wear personal protective equipment.



CAUTION

Hot or cold liquid

- Minor or moderate personal injury
- Wear personal protective equipment.



CAUTION

Hot or cold surface

- Minor or moderate personal injury
- Make sure that no one can accidentally come into contact with hot or cold surfaces.



8.1 Maintaining the product

Pump bearings and shaft seal are maintenance-free.

8.1.1 Motor bearings

Motors not fitted with grease nipples are maintenance-free.

Motors fitted with grease nipples should be lubricated with a high-temperature, lithium-based grease. See the instructions on the fan cover.

In the case of seasonal operation (motor is idle for more than 6 months of the year), we recommend to grease the motor when the pump is taken out of operation.

Depending on the ambient temperature, the motor bearings must be replaced or lubricated according to the table below. The table applies to 2-pole motors. The number of operating hours stated for bearing replacement are guidelines only.

Motor size [kW]	Bearing replacement interval [operating hours]				
	40 °C	45 °C	50 °C	55 °C	60 °C
0.37 - 0.75	18000	-	-	-	-
1.1 - 7.5	20000	15500	12500	10000	7500
Motor size [kW]	Lubrication interval [operating hours]				
	40 °C	45 °C	50 °C	55 °C	60 °C
11 - 18.5	4500	3400	2500	1700	1100
22	4000	3100	2300	1500	1000
30-55	4000	3000	2000	1500	-
75	2000	1500	1000	500	-

Intervals for 4-pole motors are twice as long as those for 2-pole motors.

If the ambient temperature is lower than 40 °C, bearings must be replaced/lubricated at the intervals mentioned under 40 °C.

8.2 Filters

Chip trays, filters, etc. must be cleaned at regular intervals to ensure a correct flow of liquid.

8.3 Periodic checks

At regular intervals, depending on the conditions and time of operation, the following checks must be made:

- Check the quantity of liquid and operating pressure.
- Check that there are no leaks.
- Check that the motor is not overheating.
- Check the tripping function of the motor-protective circuit breaker.
- Check that all controls are operating satisfactorily.

If the above checks do not reveal any abnormal operating details, no further checks are necessary. Should any faults be found, check the symptoms according to section [10. Fault finding the product](#).

8.4 Coupling adjustment

For adjustment of coupling in CRK and MTR 1s to 20, see page [399](#).

For adjustment of coupling in MTR 32, 45, 64, see page [400](#).

8.5 Contaminated products

The product will be classified as contaminated if it has been used for a liquid which is injurious to health or toxic.

If you request Grundfos to service the product, contact Grundfos with details about the liquid before returning the product for service. Otherwise, Grundfos can refuse to accept the product for service.

Any application for service must include details about the liquid.

Clean the product in the best possible way before you return it.

Costs of returning the product are to be paid by the customer.

8.6 Service documentation

Service documents and service kits are available in Grundfos Product Center (<http://product-selection.grundfos.com>).

9. Taking the product out of operation

9.1 Taking the product permanently out of operation

CAUTION

Corrosive liquids

- Death or serious personal injury
- Wear personal protective equipment.



CAUTION

Toxic liquids

- Death or serious personal injury
- Wear personal protective equipment.



CAUTION

Hot or cold liquid

- Minor or moderate personal injury
- Wear personal protective equipment.



CAUTION

Hot or cold surface

- Minor or moderate personal injury
- Make sure that no one can accidentally come into contact with hot or cold surfaces.



The product will be classified as contaminated if it has been used for a liquid which is injurious to health or toxic.

10. Fault finding the product

WARNING

Electric shock



- Death or serious personal injury
- Before starting any work on the product, make sure that the power supply has been switched off and that it cannot be accidentally switched on.

WARNING

Moving parts



- Death or serious personal injury
- Tighten the coupling screws to the correct torque.
 - Install the coupling guards securely to the pump with the screws intended for this purpose.

CAUTION

Corrosive liquids



- Death or serious personal injury
- Wear personal protective equipment.

CAUTION

Toxic liquids



- Death or serious personal injury
- Wear personal protective equipment.

CAUTION

Hot or cold liquid



- Minor or moderate personal injury
- Wear personal protective equipment.

CAUTION

Hot or cold surface



- Minor or moderate personal injury
- Make sure that no one can accidentally come into contact with hot or cold surfaces.



Fault	Cause	Remedy
1. Motor does not run when started.	a) Power supply failure.	Connect the power supply.
	b) Fuses are blown.	Replace fuses.
	c) Motor-protective circuit breaker has tripped.	Reactivate the motor-protective circuit breaker.
	d) Thermal protection has tripped.	Reactivate the thermal protection.
	e) Main contacts in motor-protective circuit breaker are not making contact or the coil is faulty.	Replace contacts or magnetic coil.
	f) Control circuit is defective.	Repair the control circuit.
	g) Motor is defective.	Replace the motor.
2. Motor-protective circuit breaker trips immediately when power supply is switched on.	a) One fuse blown/automatic circuit breaker has tripped.	Replace fuse/cut in the circuit breaker.
	b) Contacts in motor-protective circuit breaker are faulty.	Replace motor-protective circuit breaker contacts.
	c) Cable connection is loose or faulty.	Fasten or replace the cable connection.
	d) Motor winding is defective.	Replace the motor.
	e) Pump mechanically blocked.	Remove the mechanical blocking of the pump.
	f) Motor-protective circuit breaker overload setting is too low.	Set the motor-protective circuit breaker correctly.
3. Motor-protective-circuit breaker trips occasionally.	a) Motor-protective circuit breaker overload setting is too low.	Set the motor-protective circuit breaker correctly.
	b) Low voltage at peak times.	Reestablish constant power supply.
4. Motor-protective-circuit breaker has not tripped but the pump does not run.	a) Check 1 a), b), d), e) and f).	

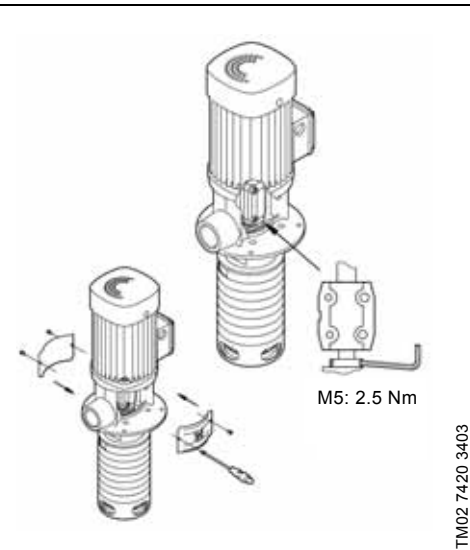
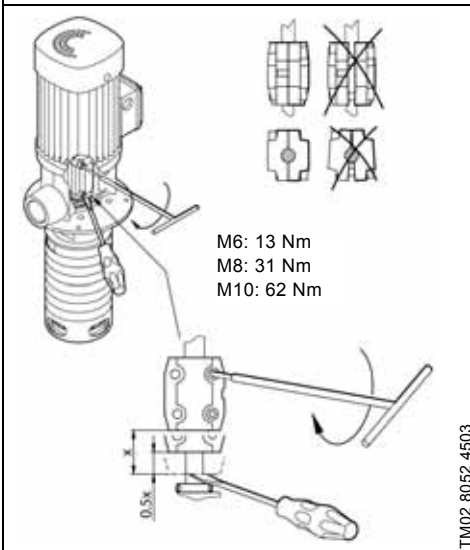
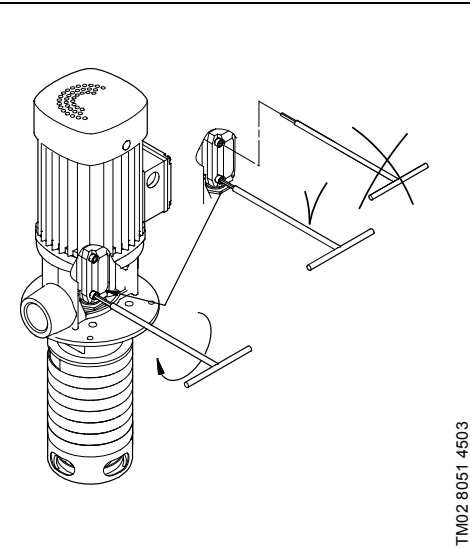
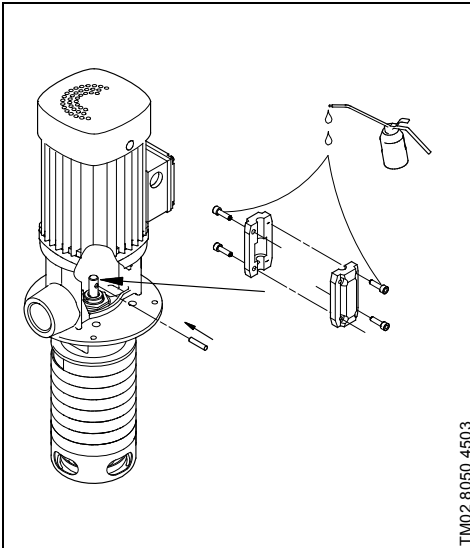
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1. Motor does not run when started.	a) Power supply failure.	Connect the power supply.
	b) Fuses are blown.	Replace fuses.
	c) Motor-protective circuit breaker has tripped.	Reactivate the motor-protective circuit breaker.
	d) Thermal protection has tripped.	Reactivate the thermal protection.
	e) Main contacts in motor-protective circuit breaker are not making contact or the coil is faulty.	Replace contacts or magnetic coil.
	f) Control circuit is defective.	Repair the control circuit.
	g) Motor is defective.	Replace the motor.
5. Pump runs but gives no liquid or pump performance is not constant.	a) Pump strainer partly blocked by impurities.	Clean the strainer.
	b) Liquid level in tank too low.	Increase the liquid level.
	c) Pump draws in air.	Check the inlet conditions.
6. Leakage in shaft seal.	a) Shaft seal is defective.	Replace the shaft seal.
7. Noise.	a) Cavitation.	Check the inlet conditions.
	b) Pump does not rotate freely (frictional resistance) because of incorrect pump shaft position.	Adjust the pump shaft.
	c) Frequency converter operation.	See section Motors supplied by Grundfos .

11. Disposing of the product

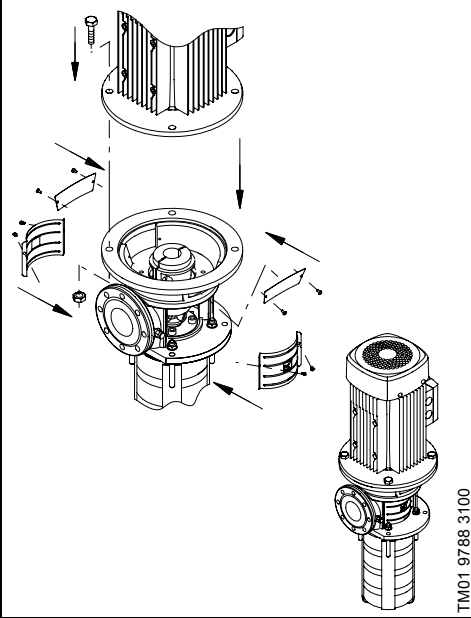
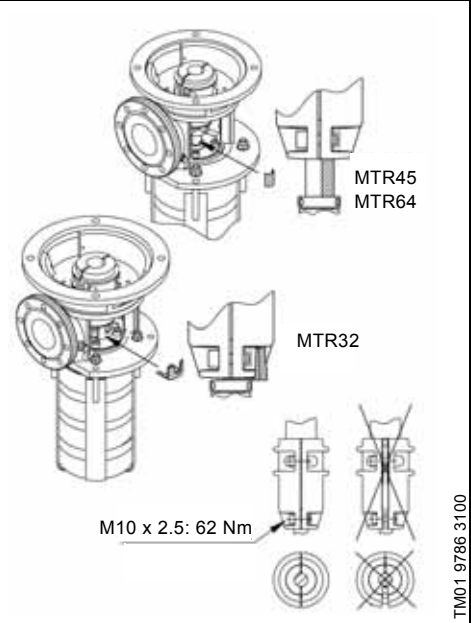
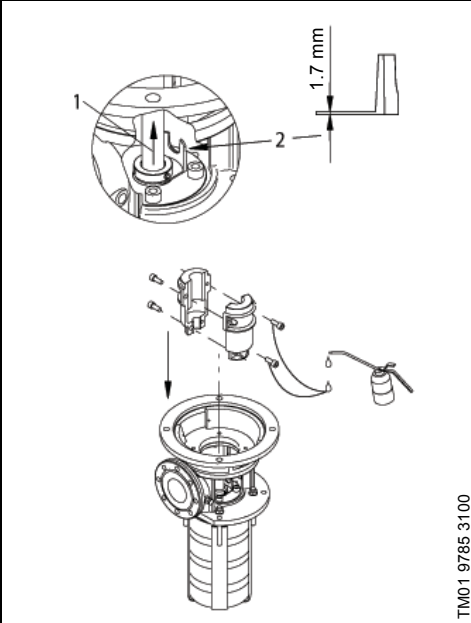
This product or parts of it must be disposed of in an environmentally sound way:

1. Use the public or private waste collection service.
2. If this is not possible, contact the nearest Grundfos company or service workshop.
3. See also end-of-life information at www.grundfos.com/product-recycling.

CRK and MTR 1s to 20



MTR 32, 45, 64



Declaration of conformity

GB: EC/EU declaration of conformity

We, Grundfos, declare under our sole responsibility that the products CRK, MTR, to which the declaration below relates, are in conformity with the Council Directives listed below on the approximation of the laws of the EC/EU member states.

CZ: Prohlášení o shodě EU

My firma Grundfos prohlašujeme na svou plnou odpovědnost, že výrobky CRK, MTR, na které se toto prohlášení vztahuje, jsou v souladu s níže uvedenými ustanoveními směrnice Rady pro sblížení právních předpisů členských států Evropského společenství.

DK: EF/EU-overensstemmelseserklæring

Vi, Grundfos, erklærer under ansvar at produkterne CRK, MTR som erklæringen nedenfor omhandler, er i overensstemmelse med Rådets direktiver der er nævnt nedenfor, om indbyrdes tilnærmelse til EF/EU-medlemsstaternes lovgivning.

ES: Declaración de conformidad de la CE/UE

Grundfos declara, bajo su exclusiva responsabilidad, que los productos CRK, MTR a los que hace referencia la siguiente declaración cumplen lo establecido por las siguientes Directivas del Consejo sobre la aproximación de las legislaciones de los Estados miembros de la CE/UE.

FR: Déclaration de conformité CE/UE

Nous, Grundfos, déclarons sous notre seule responsabilité, que les produits CRK, MTR, auxquels se réfère cette déclaration, sont conformes aux Directives du Conseil concernant le rapprochement des législations des États membres CE/UE relatives aux normes énoncées ci-dessous.

HR: EC/EU deklaracija sukladnosti

Mi, Grundfos, izjavljujemo s punom odgovornošću da su proizvodi CRK, MTR, na koja se izjava odnosi i nastavku, u skladu s dolje navedenim direktivama Vijeća o usklađivanju zakona država članica EZ-a / EU-a.

IT: Dichiarazione di conformità CE/UE

Grundfos dichiara sotto la sua esclusiva responsabilità che i prodotti CRK, MTR, ai quali si riferisce questa dichiarazione, sono conformi alle seguenti direttive del Consiglio riguardanti il ravvicinamento delle legislazioni degli Stati membri CE/UE.

LV: EK/ES atbilstības deklarācija

Sabiedrība Grundfos ar pilnu atbildību paziņo, ka produkti CRK, MTR, uz kuru attiecas tālāk redzamā deklarācija, atbilst tālāk norādītajām Padomes direktīvām par EK/ES dalībvalstu normatīvo aktu tuvināšanu.

PL: Deklaracja zgodności WE/UE

My, Grundfos, oświadczamy z pełną odpowiedzialnością, że nasze produkty CRK, MTR, których deklaracja niniejsza dotyczy, są zgodne z następującymi dyrektywami Rady w sprawie zbliżenia przepisów prawnych państw członkowskich.

RO: Declarația de conformitate CE/UE

Noi Grundfos declaram pe propria răspundere că produsele CRK, MTR, la care se referă această declarație, sunt în conformitate cu Directivele de Consiliu specificate mai jos privind armonizarea legilor statelor membre CE/UE.

RU: Декларация о соответствии нормам ЕЭС/ЕС

Мы, компания Grundfos, со всей ответственностью заявляем, что изделия CRK, MTR, к которым относится нижеприведённая декларация, соответствуют нижеприведённым Директивам Совета Евросоюза о тождественности законов стран-членов ЕЭС/ЕС.

SI: Izjava o skladnosti ES/EU

V Grundfosu s polno odgovornostjo izjavljamo, da je izdelek CRK, MTR, na katerega se spodnja izjava nanaša, v skladu s spodnjimi direktivami Sveta o približevanju zakonodaje za izenačevanje pravnih predpisov držav članic ES/EU.

BG: Декларация за съответствие на ЕС/ЕО

Ние, фирма Grundfos, заявяваме с пълна отговорност, че продуктите CRK, MTR, за които се отнася настоящата декларация, отговарят на следните директиви на Съвета за уеднавяване на правните разпоредби на държавите-членки на ЕС/ЕО.

DE: EG-/EU-Konformitätserklärung

Wir, Grundfos, erklären in alleiniger Verantwortung, dass die Produkte CRK, MTR, auf die sich diese Erklärung bezieht, mit den folgenden Richtlinien des Rates zur Angleichung der Rechtsvorschriften der EG-/EU-Mitgliedsstaaten übereinstimmen.

EE: EÜ / ELi vastavusdeklaratsioon

Meie, Grundfos, kinnitame ja kanname ainuiskulist vastutust selle eest, et toode CRK, MTR, mille kohta all olev deklaratsioon käib, on kooskõlas Nõukogu Direktiividega, mis on nimetatud all pool vastavalt vastuvõetud õigusaktidele ühtlustamise kohta EÜ / EL liikmesriikides.

FI: EY/EU-vaatimustenmukaisuusvakuutus

Grundfos vakuuttaa omalla vastuullaan, että tuotteet CRK, MTR, joita tämä vakuutus koskee, ovat EY/EU:n jäsenvaltioiden lainsäädännön lähentämiseen tähtäviin Euroopan neuvoston direktiivien vaatimusten mukaisia seuraavasti.

GR: Δήλωση συμμόρφωσης ΕΚ/ΕΕ

Εμείς, η Grundfos, δηλώνουμε με αποκλειστική δική μας ευθύνη ότι τα προϊόντα CRK, MTR, στα οποία αναφέρεται η παρακάτω δήλωση, συμμορφώνονται με τις παρακάτω Οδηγίες του Συμβουλίου περί προσέγγισης των νομοθεσιών των κρατών μελών της ΕΚ/ΕΕ.

HU: EC/EU megfeleléségi nyilatkozat

Mi, a Grundfos vállalat, teljes felelősséggel kijelentjük, hogy a(z) CRK, MTR termékek, amelyre az alábbi nyilatkozat vonatkozik, megfelelnek az Európai Közösség/Európai Unió tagállamainak jogi irányelveit összefoglaló tanács alábbi előírásainak.

LT: EB/ES atitikties deklaracija

Mes, Grundfos, su visa atsakomybe pareiškiamo, kad produktai CRK, MTR, kuriems skirta ši deklaracija, atitinka žemiau nurodytas Tarybos Direktyvas dėl EB/ES šalių narių įstatymų suderinimo.

NL: EG/EU-conformiteitsverklaring

Wij, Grundfos, verklaren geheel onder eigen verantwoordelijkheid dat de producten CRK, MTR, waarop de onderstaande verklaring betrekking heeft, in overeenstemming zijn met de onderstaande Richtlijnen van de Raad inzake de onderlinge aanpassing van de wetgeving van de EG-/EU-lidstaten.

PT: Declaração de conformidade CE/UE

A Grundfos declara sob sua única responsabilidade que os produtos CRK, MTR, aos quais diz respeito a declaração abaixo, estão em conformidade com as Directivas do Conselho sobre a aproximação das legislações dos Estados Membros da CE/UE.

RS: Deklaracija o uskladenosti EC/EU

Mi, kompanija Grundfos, izjavljujemo pod punom vlastitom odgovornošću da je proizvod CRK, MTR, na koji se odnosi deklaracija ispod, u skladu sa dole prikazanim direktivama Saveta za usklađivanje zakona država članica EC/EU.

SE: EG/EU-försäkran om överensstämmelse

Vi, Grundfos, försäkrar under ansvar att produkterna CRK, MTR, som omfattas av nedanstående försäkran, är i överensstämmelse med de råddirektiv om inbördes närmande till EG/EU-medlemsstaternas lagstiftning som listas nedan.

SK: Prehlásenie o zhode s EC/EU

My, spoločnosť Grundfos, vyhlasujeme na svoju plnú zodpovednosť, že produkty CRK, MTR na ktoré sa vyhlásenie uvedené nižšie vzťahuje, sú v súlade s ustanoveniami nižšie uvedených smerníc Rady pre zblíženie právnych predpisov členských štátov Európskeho spoločenstva/EÚ.

TR: EC/AB uygunluk bildirgesi

Grundfos olarak, aşağıdaki bildirim konusu olan CRK, MTR ürünlerinin, EC/AB Üye ülkelerinin direktiflerinin yakınlştırılmasıyla ilgili durumun aşağıdaki Konsey Direktifleriyle uyumlu olduğunu ve bununla ilgili olarak tüm sorumluluğun bize ait olduğunu beyan ederiz.

BS: Izjava o usklađenosti EC/EU

Mi, kompanija Grundfos, izjavljujemo pod vlastitom odgovornošću da je proizvod CRK, MTR, na koji se odnosi izjava ispod, u skladu sa niže prikazanim direktivama Vijeća o usklađivanju zakona država članica EC/EU.

UA: Декларація відповідності директивам ЕС/ЕУ

Ми, компанія Grundfos, під нашу одноосібну відповідальність заявляємо, що вироби CRK, MTR, до яких відноситься нижченаведена декларація, відповідають директивам ЕС/ЕУ, переліченим нижче, щодо тотожності законів країн-членів ЄС.

- Machinery Directive (2006/42/EC).
Standard used: EN 809:1998, A1:2009
- Ecodesign Directive (2009/125/EC).
Electric motors:
Commission Regulation No 640/2009.
Applies only to three-phase Grundfos motors marked IE2 or IE3.
See motor nameplate.
Standard used: EN 60034-30:2009.
- Water pumps:
Commission Regulation No 547/2012.
Applies only to water pumps marked with the minimum efficiency index MEI. See pump nameplate.

This EC/EU declaration of conformity is only valid when published as part of the Grundfos installation and operating instructions (publication number 96496966).

Additional directives and standards effective from 22 July 2019:

- RoHS Directives: 2011/65/EU and 2015/863/EU
Standard used: EN 50581:2012

Bjerringbro, 17 December 2018



Erik Andersen
Senior Manager
GRUNDFOS Holding A/S
Poul Due Jensens Vej 7
8850 Bjerringbro, Denmark

Manufacturer and person empowered to sign the EC/EU declaration of conformity

RUS

MTR

Руководство по эксплуатации

Руководство по эксплуатации на данное изделие является составным и включает в себя несколько частей:

Часть 1: настоящее «Руководство по эксплуатации».

Часть 2: электронная часть «Паспорт. Руководство по монтажу и эксплуатации» размещенная на сайте компании Грундфос. Перейдите по ссылке, указанной в конце документа.

Часть 3: информация о сроке изготовления, размещенная на фирменной табличке изделия.

Сведения о сертификации:

Насосы типа MTR сертифицированы на соответствие требованиям Технических регламентов Таможенного союза: ТР ТС 004/2011 «О безопасности низковольтного оборудования»; ТР ТС 010/2011 «О безопасности машин и оборудования»; ТР ТС 020/2011 «Электромагнитная совместимость технических средств».

KAZ

MTR

Пайдалану бойынша нұсқаулық

Атаулы өнімге арналған пайдалану бойынша нұсқаулық құрамалы болып келеді және келесі бөлімдерден тұрады:

1 бөлім: атаулы «Пайдалану бойынша нұсқаулық»

2 бөлім: Грундфос компаниясының сайтында орналасқан электронды бөлім «Төлқұжат, Құрастыру және пайдалану бойынша нұсқаулық». Құжат соңында көрсетілген сілтеме арқылы өтіңіз.

3 бөлім: өнімнің фирмалық тақташасында орналасқан шығарылған уақыты жөніндегі мәлімет

Сертификаттау туралы ақпарат:

MTR типті сорғылары «Төмен вольтты жабдықтардың қауіпсіздігі туралы» (ТР ТС 004/2011), «Машиналар және жабдықтар қауіпсіздігі туралы» (ТР ТС 010/2011) «Техникалық заттардың электрлі магниттік сәйкестілігі» (ТР ТС 020/2011) Кеден Одағының техникалық регламенттерінің талаптарына сәйкес сертификатталды.

KG

MTR

Пайдалануу боюнча колдонмо

Аталган жабдууну пайдалануу боюнча колдонмо курамдык жана өзүнө бир нече бөлүкчөнү камтыйт:

1-Бөлүк: «Пайдалануу боюнча колдонмо»

2-Бөлүк: «Паспорт. Пайдалануу жана монтаж боюнча колдонмо» электрондук бөлүгү Грундфос компаниянын сайтында жайгашкан. Документтин аягында көрсөтүлгөн шилтемеге кайрылыңыз.

3-Бөлүк: жабдуунун фирмалык тактасында жайгашкан даярдоо мөөнөтү тууралуу маалымат.

Шайкештик жөнүндө декларация

MTR түрүндөгү соргучтар Бажы Биримдиктин Техникалык регламенттин талаптарына ылайыктуу тастыкталган:

ТР ТБ 004/2011 «Төмөн вольттук жабдуунун коопсуздугу жөнүндө»; ТР ТБ 010/2011 «Жабдуу жана машиналардын коопсуздугу жөнүндө»; ТР ТБ 020/2011 «Техникалык каражаттардын электрмагниттик шайкештиги».

ARM

MTR

Հահագործման ձեռնարկ

Տվյալ սարքավորման շահագործման ձեռնարկը բաղկացած է մի քանի մասերից.

Մաս 1. սույն «Հահագործման ձեռնարկ»:

Մաս 2. էլեկտրոնային մաս. այն է՝ «Անձնագիր: Մոնտաժման և

շահագործման ձեռնարկ» տեղադրված «Գրունտֆոս». Անցեք փաստաթղթի վերջում նշված հղումով.

Մաս 3. տեղեկություն արտադրման ամսաթվի վերաբերյալ՝ նշված սարքավորման պիտակի վրա:

Տեղեկություններ հավաստագրման մասին՝

MTR տիպի պոմպերը սերտիֆիկացված են համաձայն Մաքսային Միության տեխնիկական կանոնակարգի պահանջների՝ TP TC 004/2011 «Ցածրավոլտ սարքավորումների վերաբերյալ», TP TC 010/2011 «Մեքենաների և սարքավորումների անվտանգության վերաբերյալ» ; TP TC 020/2011 «Տեխնիկական միջոցների էլեկտրամագնիսական համատեղելիության վերաբերյալ»:



<http://net.grundfos.com/qr/i/99310687>

10000142896	0717
ECM: 1213063	

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